College of Engineering

Department of Software Engineering

Internet Programming I (SWEG3107)

Musical Instrument Store Documentation

Group Members

1. Kidst Nigus - ETS1677/16
2. Kirubel Lemma - ETS0837/16
3. Nardos Anteneh - ETS1076/16
4. Nazrawit Tewodros - ETS1109/16
5. Samuel Abebe - ETS1219/16

Submitted to: Mr. Jerusalem F.

Submission date: 19/01/2026

TABLE OF CONTENTS

Introduction

For any business in the modern age having a digital presence has become more and more important as time goes on. This is also true for the musical instrument market because musicians and hobbyists want a way to easily browse, select and purchase their instruments online from the comfort of their homes. Moreover, due to the expensive prices of musical instruments people are also looking for ways to minimize their costs and still get to use the instruments they love. This brings us to the website this document is about, “The Fine Tune”.

This document presents the design and development of “The Fine Tune”, an online musical instrument store that bridges the gap between the physical store and an online presence. The project explores how a thoughtfully designed e-commerce platform can enhance accessibility to musical instruments through both purchasing and rental services. By integrating technical functionality with business logic and user-centered design, this project aims to provide a solution for musicians of varying skill levels, financial capacities, and creative aspirations.

The following sections outline the project’s context, motivation, problem statement, objectives, beneficiaries, significance, feasibility and system requirements specification, offering a structured overview of the system and the value it seeks to deliver within the modern digital marketplace.

Context

In current times to be competitive in the market just having a physical store space is not enough. A whole world of e-commerce exists in the digital world where customers can find and purchase goods and services from different businesses. In this regard any store would benefit from having a space in this digital world and expanding their reach to increase the amount of customers they can have. A musical instrument store is one of the many businesses that can benefit from this. By having this digital place the store can showcase its items to potential customers and even rent or sell its items online. This project aims to build a website for a musical instrument store to provide this digital space.

Motivation

Our motivation to build ‘Fine tunes’, an online music store, stems from the creative curiosity and interest we have towards music and the beautiful process behind it. This is an interest we share with many other peers, in fact data from 2024 shows that 83% of young people listened to music in the past week, while 71% consider music to be a fundamental part of their identity. Roughly 83% of teenagers listen to music for about 2 hours daily and 60% of students listen to music while studying. And this data is evident in the daily lives of many young adults. This assignment allowed us to integrate two things we care about, music and coding, into something tangible, something we would be proud to show our peers.

Beyond personal passion, we were motivated by the opportunity to tackle a project with both coding and business complexity to elevate our skill sets. The inclusion of the rental model isn't just a mere feature, to be honest, but a bridge that can help so many of us cross the huge gap created by the expensive nature of these instruments. Same goes for the accessibility features like the left and right hand option in our product details page, these things are the little details that come together to make this project ours. This project presented a chance to show holistic understanding of concepts in website development, and beyond that, the business side of this, the logic and the necessary details in customer service in this type of platform to make it that much more accessible.

We were particularly drawn to the possibility of designing something as rich as a music domain. Unlike generic E-Commerce projects, this project encourages creativity, bold aesthetic, and the freedom to experiment with the theme and interactive nature. This allowed us to reflect our passion for this project and stir up the passions of the musicians that visit our website. The biggest puzzle was how to make functionality intuitive and yet seamless, and the combination of our creative freedom paired with technical depth required for this project made the process that much more invaluable.

Problem Statement

Existing online marketplace places for musical instruments often suffer from generic, uninspiring designs that fail to capture the passion and creativity of music and musicianship. Furthermore, they tend to be fragmented, like specific brands or only specific instruments (like a guitar only stores), or solely work on sales and rentals. This forces musicians to go through multiple bland websites to meet their need creating an unsatisfactory expiriance. Our project integrates all this seamlessly making the experience of the user shopping or renting that much more convenient.

Musicians don't just shop for a 'product', they are looking for a tool that brings their inner tunes out in a way they find confidence in, to a musician his tool is his work, it is what defines what he does. This requires exploring the vast ecosystem of instruments like Fendor and Gibson and Yamaha across categories like guitar, pianos, speakers, keyboards. Our customers need a single destination that facilitates this process. Their needs are as financial as they are creative, a student might dream of a Yamaha PSR- E483, alas what they can afford might just be rent. A band might want a vintage set of instruments just for one show, they might now feel completed to buy it, which is why they come to us, they might need a set of speakers, they need to buy that, which is why they come to us as well. We seamlessly assemble a smooth experience from searching to acquisition .

As evidently presented, there is a significant gap in the market. And the solution seems to be just as evident, a unified platform that combines curated, multi-brand inventory with an emotionally resonant, visually appealing design and a flexible dual-transaction model.

Objective

General Objective

Creating a website that can act like a digital location for a musical instrument store, where customers can get information about the different items inside of the musical instrument store, and rent out or purchase musical instruments that are available in the store.

Specific Objectives

Creating a musical instrument store website that:

- displays a catalog of the available stock at the store to users.
- displays detailed information about the available stock at the store to users.
- allows users to search for specific items in stock at the store.
- allows users to rent items in stock at the store.
- allows users to purchase items in stock at the store.
- displays a cart page that shows the items currently in the cart of the user.
- displays a purchase summary with a detailed list of total price, taxes and other important information of the items in the user's cart.
- allows users to enter their personal information and payment method to complete a purchase.
- allows users to create a store account for purchasing and renting items from the website.
- allows users to sign in to the website with their store accounts.

Functional Requirements

The website will be able to perform the following actions:

FR1: Display a catalog of the different items in stock at the musical instrument store.

FR2: Display information about the different items in stock at the musical instrument store.

FR3: Allow searching for specific items in stock at the musical instrument store.

FR4: Allow renting of items in stock at the musical instrument store.

FR5: Allow purchasing of items in stock at the musical instrument store.

FR6: Display a cart showing the number of items currently in the cart.

FR7: Display a purchase summary containing a detailed sum of the prices of the items in the cart.

FR8: Allow users to input their personal information and payment method to complete a purchase.

FR9: Allow users to make a store account.

FR10: Allow users to sign in with their store accounts.

Beneficiaries of the System

In this section we will discuss the primary, secondary and indirect beneficiaries of this website.

Beneficiary Group	Key problems addressed	Platform features used	Primary value gained
Student & Aspiring Musicians	Expensive price for quality products	Rental system in progress and accessible to buy.	Lowered financial barrier to entry; risk-free experimentation with equipment
Gigging Professionals	Needs a specialized hear for a one tike projects	Short term rental option,	Same as above
Enthusiasts	Limited access to diverse equipment, inspired but unable to access instruments	Visually engaging UI/UX, you can rent before you buy	Enhanced enjoyment of browsing; safe exploration of new instrument categories

Figure 1. Primary beneficiary

Beneficiary	Traditional Limitations	Platform Benefits
Independent Music Stores	Limited geographical reach	Expanded digital footprint
Instrument Manufacturers	Disconnected customer feedback; limited demo opportunities	Real-time market data, brand exposure
Music Educators & Institutions	Budget constraints, equipment maintenance	Flexible rental terms for academic calendars; bulk rental discounts

Figure 2. Secondary beneficiaries.

Beyond the direct market uses to the participants, the website brings about other nuanced benefits as well. For example the societal benefit it has to make music accessible to the general public will raise a generation inclined towards music without the intimidation that comes with having an instrument. It creates cultural richness and a cycle that keeps our platform thriving. This cycle where the rental revenue funds inventory expansion, which draws more users, which in turn strengthens network effects. This self-reinforcing loop drives the platform's sustainable growth and continuously multiplies value for the music community, solidifying its role as a resilient solution to core market failures.

Significance

This project is important because it confronts real world musician problems, the hustles of visiting many unappealing websites with the uncertainty is what is the perfect, and then the fragmented-ness of the whole process. 'Fine tunes' is designed to bring a practical solution that makes buying and renting an instrument a flexible process.

The work shows clear, hands-on skill in creating a complex website with business and technical layers with interactive features that actually work for the end user. It proves our ability to handle real-world challenges like designing an engaging interface, structuring logical code, and identifying real problems we can relate too. This makes the project a standout piece of practical work that speaks directly to the skills gained through Internet programming (1) course.

Feasibility Analysis

The feasibility analysis evaluates whether the proposed The Fine Tune online musical instrument store can be realistically developed and deployed within the given constraints. This assessment considers the system's technical, operational and economic feasibility in relation to the defined functional requirements (FR1–FR10).

Technical Feasibility

The system is technically feasible, as all functional requirements can be implemented using widely adopted and well-supported web development technologies. Features such as product catalog display (FR1), item detail pages (FR2), and search functionality (FR3) can be achieved through structured databases, server-side logic, and responsive front-end frameworks.

Rental (FR4) and purchase (FR5) functionalities can be handled through transaction-based workflows that update inventory availability in real time. Cart management (FR6) and purchase summaries (FR7) rely on session handling and dynamic price calculations, which are standard practices in e-commerce systems.

User account creation (FR9) and authentication (FR10) can be implemented securely using encrypted password storage and authentication mechanisms. Payment data input (FR8) can be supported through form validation and integration with secure payment gateways. Overall, the project does not require experimental or proprietary technologies, making it technically achievable.

Operational Feasibility

From an operational perspective, the system is designed to be intuitive and accessible to users with varying levels of technical experience. Browsing instruments, searching for specific items, and viewing product details closely mimic familiar e-commerce workflows, reducing the learning curve.

The inclusion of both rental and purchase options addresses real user needs and enhances usability for students, professionals, and enthusiasts alike. Account creation and login features allow users to manage their activities securely, while the cart and checkout process ensures a smooth end-to-end transaction flow. Store administrators can manage inventory and availability without complex operational overhead, making the system practical for real-world use.

Economic Feasibility

The project is economically feasible due to its reliance on cost-effective or free development tools, frameworks, and databases. No specialized hardware is required beyond standard development and hosting environments.

From a business perspective, the dual rental-and-purchase model maximizes revenue potential by catering to users who can not afford outright purchases. Rentals allow high-value instruments to generate recurring income, while purchases support long-term profitability. This flexible model lowers financial barriers for customers while ensuring sustainable platform growth.

System Requirement Specification

Existing System Overview

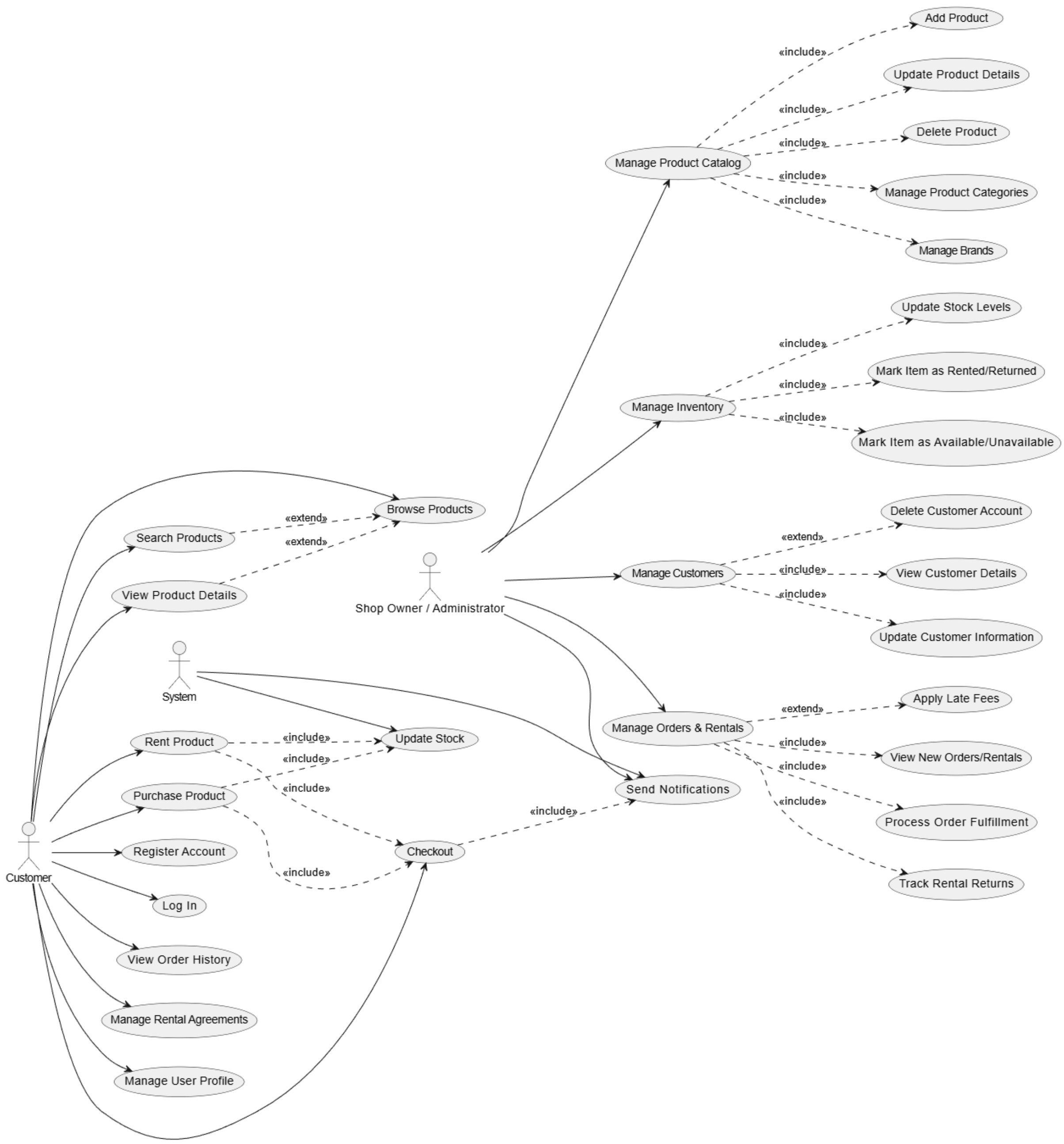
Most music and equipment shops in Ethiopia, particularly in areas like Addis Ababa (Bole, Piazza, or Megenagna), rely almost entirely on physical walk-ins. There is no central digital hub where a customer can see what is available across the city. If a musician is looking for a specific brand of guitar or a specific model of studio monitor, they usually have to physically travel from shop to shop, or spend hours making phone calls to different owners to ask, "Is it in stock?" This "foot-on-the-ground" method is time-consuming and inefficient.

Behind the counter, the existing system is often very manual. Many shops still use paper-based ledgers or, at best, a simple Excel sheet that is only updated at the end of the day or week. This means that the "available" stock is often inaccurate. A shop owner might tell a customer over the phone that a keyboard is available, only for the customer to arrive an hour later and find it was sold to someone else five minutes prior. There is no real-time sync between the shelf and a database, leading to missed sales and frustrated buyers.

While many Ethiopian shops use social media platforms like Telegram, TikTok, or Facebook Marketplace to post pictures of their items, these are not "systems." They are just digital notice boards. There is no way to filter by brand, no way to see real-time quantities, and no way to process a secure transaction. Furthermore, these social media posts quickly get buried under new content, making it hard for customers to find older but still available products. The "centralized place" for brand-specific searching simply does not exist yet.

Renting equipment (like sound systems for weddings, or cameras/instruments for music videos) is very common in Ethiopia, but the current system is highly informal. It usually depends on "who you know." A customer has to find a shop willing to rent, and the "contract" is often just a verbal agreement or a handwritten note. Because there is no formal database to track rental periods, return dates, or deposits, shop owners take a huge risk of losing their equipment, and customers face inconsistent pricing. There is no transparent, automated system that calculates rental fees based on duration.

Use Case Diagram



Problems of Existing System

The business logic of the existing system is very traditional and relies on trust in vendors that provide a complete check and balance. By creating a better system to deal with the business we can increase revenue and claim business opportunities in a better and well organized manner.

Proposed System

In the current retail landscape, customers often find themselves frustrated by the fragmented nature of online shopping. If a user is looking for high-quality products from specific, trusted brands, they usually have to jump between dozens of different websites, compare shifting prices, and deal with varying shipping policies. Our proposed solution aims to solve this "search fatigue" by creating a sophisticated, centralized system that brings products from various premier brands together into one single, high-performance platform. By providing a one-stop-shop experience, we are not just selling products; we are providing a simplified lifestyle for the modern consumer who values their time as much as their money.

This project is far more than just a standalone website; it is designed to function as a direct digital extension of an actual physical store. One of the biggest issues with standard e-commerce is the "ghost inventory" problem, where an item is listed as available online but is actually out of stock in the warehouse. Because our system is built to work hand-in-hand with a real-world stock management database, the inventory is always accurate. When an item is touched, moved, or sold in the physical shop, the webstore reflects that change immediately. This synergy between the physical and the digital ensures that the business operates as one cohesive unit, providing a reliable "omnichannel" experience that customers can trust.

What truly sets this platform apart from the competition is our dedicated rental module. While most traditional retail stores focus purely on the "point-of-sale," we recognize that modern consumers are increasingly interested in the "access-over-ownership" model. There are many high-end brand products that a customer might need for a one-time event, a professional project, or a short-term trial. By offering a rental option—something that the vast majority of competitors completely ignore—we open up a massive new secondary revenue stream. This allows us to serve a wider demographic of customers who want premium quality without the commitment of a full-price purchase, making luxury and high-performance gear more accessible to everyone.

Use Case Diagram